WIEGNITE 3.0

Idea Title: PublicRush-Predict Before Prevent

Team Name: TestCase

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Problem statement and target users

The Problem

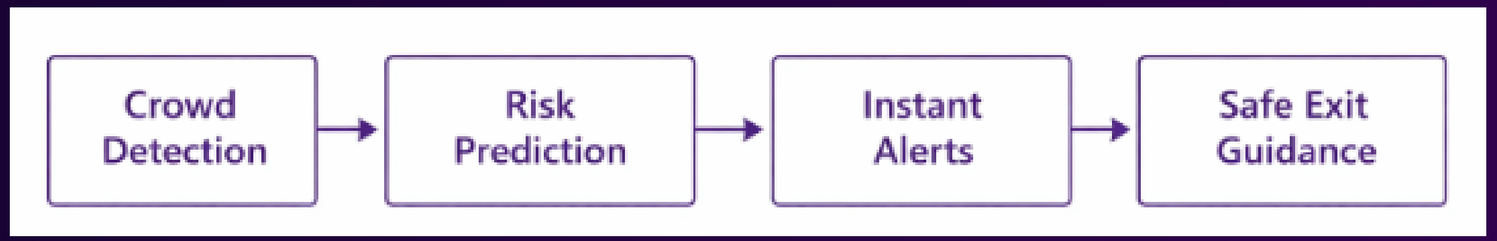
- Sudden crowd surges at public events can turn deadly within minutes.
- Authorities lack real-time visibility into crowd density.
- Most crowd control systems are reactive, not predictive.
- Manual monitoring fails in high-density situations.
- Absence of early warnings leads to panic, stampedes, and preventable casualties.

Target Users

- Law Enforcement & Disaster Response Teams - Require early warnings to avoid crowd emergencies.
- Event Organizers & Venue Managers - Need real-time monitoring to handle large crowds safely.
- Smart City Authorities - Look for scalable solutions to enhance public safety infrastructure.
- Transport Hubs & Stadium Operators - Manage high-density crowds with limited predictive capabilities.

Proposed solution

PublicRush is an AI-powered crowd intelligence system that predicts dangerous crowd buildup and prevents stampedes before they occur.



- The flowchart shows how PublicRush prevents crowd disasters through early detection and action. AI first monitors crowd density (Crowd Detection) and predicts potential risks (Risk Prediction). The system then sends Instant Alerts to authorities, followed by Safe Exit Guidance that directs people to less congested routes, ensuring safer and more controlled evacuations.

Values Delivered:

- Enables early intervention
- Guides crowds safely during emergencies
- Reduces panic and casualties
- Creates safer, smarter public spaces

Key uniqueness or innovation

Feature	Traditional Crowd Management	PublicRush
Approach	Reactive (acts after danger)	Predictive (detects risk early)
Monitoring	Manual supervision	AI-powered real-time analysis
Alerts	Delayed communication	Instant automated alerts
Public Guidance	Minimal or none	Mobile navigation to safe exits
Decision Making	Guesswork under pressure	Data-driven risk scoring

- **Predictive Crowd Pressure Mapping — Anticipates risk by analyzing movement patterns, not just crowd size.**
- **Intelligent Escape Routing — Guides people toward optimal exits, preventing bottlenecks and chaos.**

Proposed tech stack

- Python and Libraries: Uses OpenCV and YOLO for real-time crowd detection, TensorFlow/PyTorch for predictive risk modeling, and NumPy/Pandas for fast data processing.

- NodeJS and MongoDB: NodeJS enables high-speed real-time communication, while MongoDB stores crowd data, risk scores, and alert logs for scalable monitoring.

- React and Flutter: React powers the live authority dashboard, while Flutter supports a cross-platform mobile app that sends alerts and guides users to safe exits.

Live Demo: crowdrush.netlify.app